MODULE 02

Biomechanics & Movement Science

Core Learning · Written Guide + Quiz

WHY IT MATTERS

Biomechanics uses physics and engineering concepts to describe how forces act on and within the body during movement. As a climbing coach, understanding biomechanics allows you to identify inefficient movement, predict injury risk, and provide precise technical feedback. Every coaching cue you give — 'push through your feet', 'drop your hip', 'keep your elbow in' — is rooted in biomechanical principles.

Learning Objectives

By the end of this module, you will be able to:

- Identify the three planes of motion and axes of rotation and relate them to climbing movements
- Explain the concept of levers, torque, and force production in the context of the human body
- Describe and coach the seven fundamental movement patterns relevant to climbing
- Conduct a basic postural assessment and identify common movement dysfunctions in climbers
- Apply the principles of stability, mobility, and motor learning to coaching practice

Branches of Biomechanics

Branch	Description
Statics	Studies bodies in constant motion with no acceleration — foundational to postural and structural analysis.
Dynamics	Involves acceleration; studies systems with changing velocity including running, jumping, and sport-specific movement.
Kinetics	The study of forces causing motion ($F = ma$, torque, GRF).
Kinematics	The description and appearance of motion, independent of forces — angles, velocity, displacement.

Section 1: Planes of Motion & Axes of Rotation

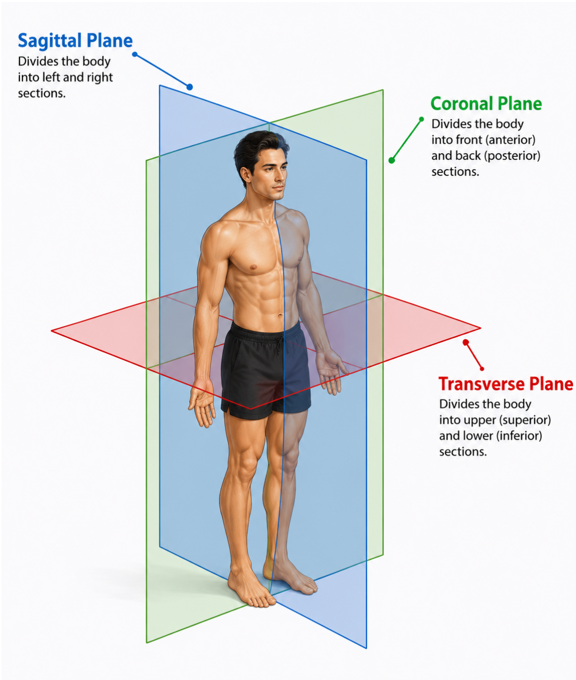
Human movement occurs in three-dimensional space. To describe and analyse movement precisely, we use three anatomical planes and their corresponding axes of rotation. A solid grasp of these concepts allows coaches to describe movements clearly and identify when a climber is compensating outside their optimal movement plane.

1.1 The Three Planes

Plane	Divides the Body Into	Example Movements
Sagittal Plane	Left and right halves	Pull-up, squat, heel raise, bicep curl — forward/backward movements
Frontal (Coronal) Plane	Front and back halves	Side step, lateral raise, hip abduction — side-to-side movements

Plane	Divides the Body Into	Example Movements
Transverse (Horizontal) Plane	Upper and lower halves	Hip rotation, shoulder rotation, twisting drop-knee — rotational movements

Climbing is a highly multi-planar activity. A single sequence on a route may involve a sagittal plane pull, a transverse plane hip rotation (drop-knee), and a frontal plane reach. Coaches who understand planes of motion can quickly identify when a move is failing because the athlete is attempting it in the wrong plane.

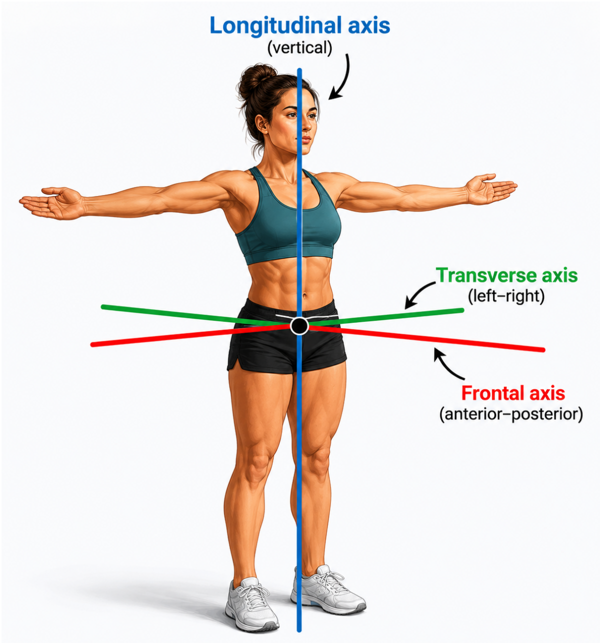


The sagittal, coronal, and transverse anatomical planes

1.2 Axes of Rotation

Each plane has a corresponding axis around which rotational movements occur:

- **Transverse Axis (Left-Right axis)** — perpendicular to the sagittal plane; allows flexion and extension around it (e.g. elbow bend)
- **Frontal (Anteroposterior) axis** — perpendicular to the frontal (coronal) plane; allows abduction and adduction around it (e.g. leg raising to the side)
- **Longitudinal (Vertical) axis** — perpendicular to the transverse plane; allows rotation around it (e.g. trunk twist for a cross-through)



The longitudinal, transverse, and frontal axes of rotation

1.3 Anatomical Terminology for Movement

Term	Definition	Climbing Example
Flexion	Decreasing the angle at a joint	Pulling into a lock-off (elbow flexion)
Extension	Increasing the angle at a joint	Straightening the arm on a rest position
Abduction	Moving a limb away from the midline	Raising the leg to the side for a high foothold
Adduction	Moving a limb toward the midline	Drawing the arm in during a lat-dominant move
Rotation (Internal/External)	Rotating a limb along its long axis	Shoulder external rotation on a pinch hold
Pronation / Supination	Forearm rotation (palm down / palm up)	Switching grip orientation on a sloper vs. undercling
Dorsiflexion / Plantarflexion	Ankle up / ankle down	Heel hook (plantarflexed); toe hook (dorsiflexed)

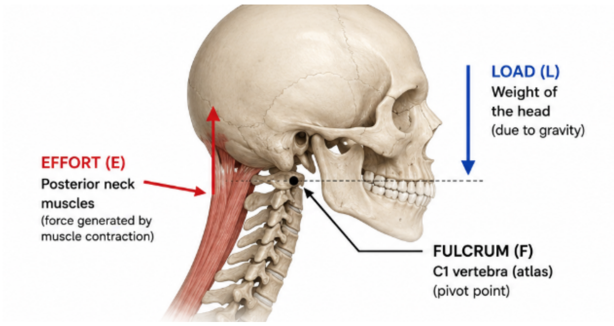
Section 2: Basic Biomechanical Concepts

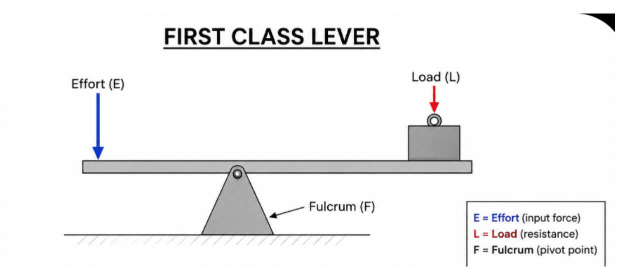
2.1 Lever Systems

Biomechanics at the joint level is governed by lever mechanics. Every joint in the body acts as a lever system, with bones as levers, joints as fulcrums, and muscles providing the force. A lever consists of a rigid bar (bone), a fulcrum (joint), an effort force (muscle), and a resistance force (weight or external load). There are three classes of lever, all present in the human body:

First Class — Fulcrum between effort and resistance

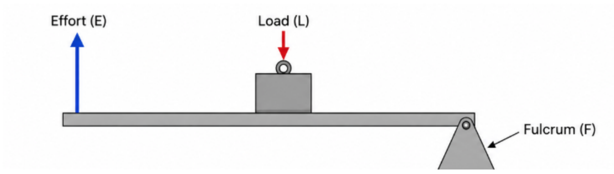
Example: the atlanto-occipital joint (head nodding) — the neck muscles pull on the back of the skull to balance the face/skull weight in front.

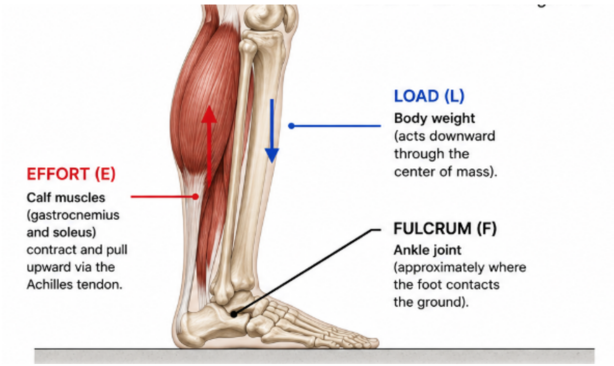




Second Class — Resistance between fulcrum and effort

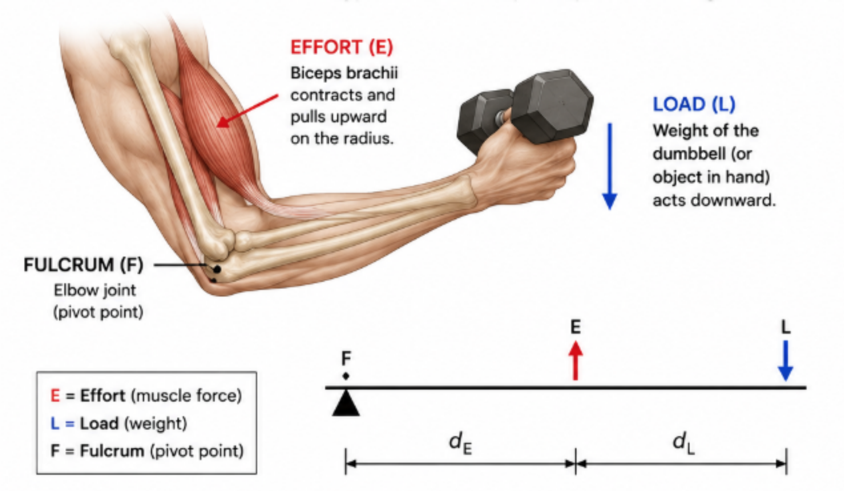
Example: calf raise — the ball of the foot is the fulcrum, body weight the resistance, and the calf muscle the effort. Mechanically advantageous (force magnified).





Third Class — Effort between fulcrum and resistance

MOST COMMON in the body. Example: bicep curl — elbow joint is fulcrum, biceps pulls just below the elbow, and the dumbbell is at the hand. Mechanically disadvantageous (speed favoured over force).



The third class lever at the elbow during a bicep curl

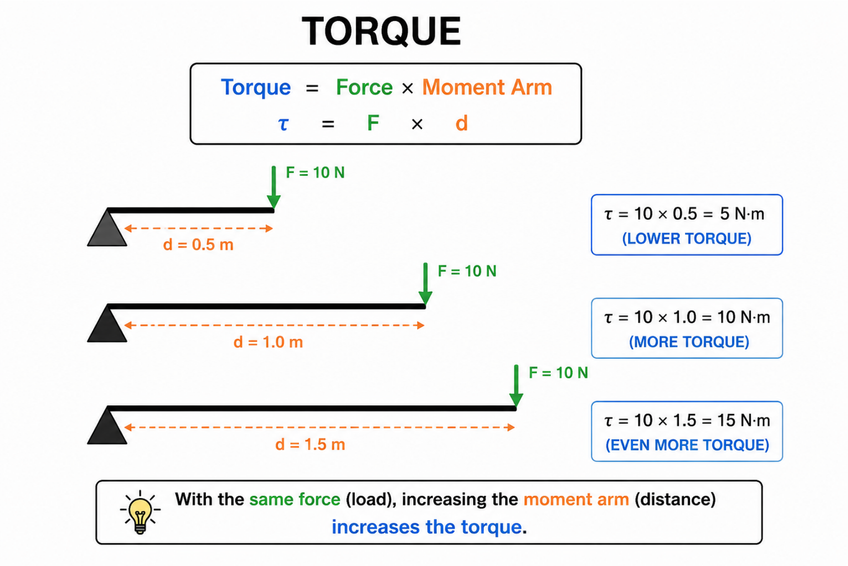
COACHING INSIGHT

Because most muscles operate as third-class levers, the body must generate far more force than the load being lifted. This is why body position on the wall matters so much: moving the centre of mass closer to the wall reduces the effect of gravity and torque on the climber's body and decreases the muscular effort required to maintain position. Teaching climbers to 'stay close to the wall' is a biomechanical principle.

2.2 Torque

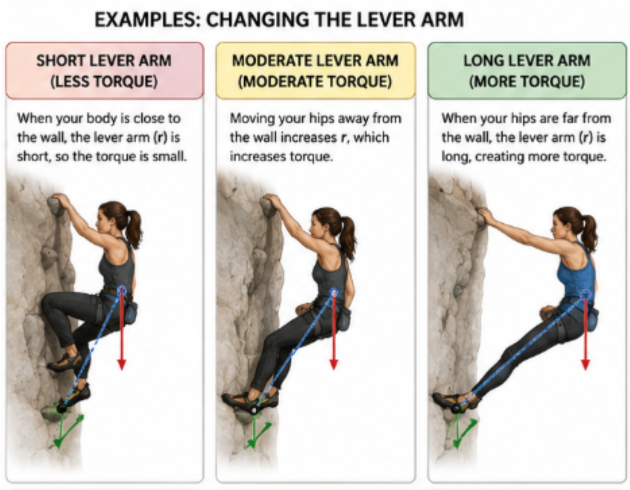
Torque (moment of force) is the rotational force applied around an axis. It is calculated as:

Torque = Force x Moment Arm (perpendicular distance from axis to line of force)



Torque increases as the moment arm increases for the same force

In practical coaching terms: the further a body part is from a joint axis, the greater the torque — and therefore the greater the muscular demand.



Short, moderate, and long lever arm examples on the climbing wall

2.3 Ground Reaction Forces and the Wall

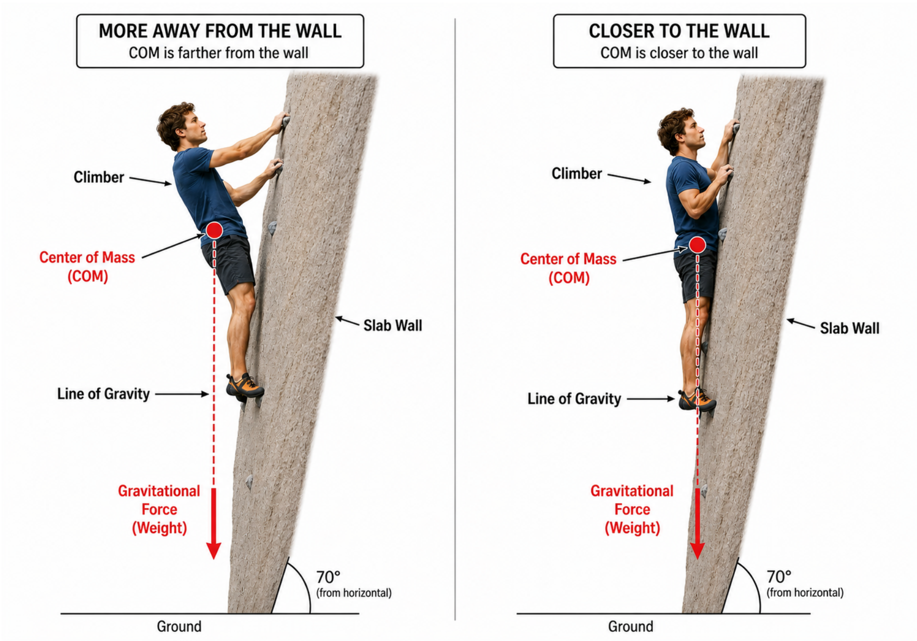
Newton's Third Law states that every action has an equal and opposite reaction. When a climber pushes down through a foothold, the wall pushes back with equal force. Understanding force vectors helps coaches explain why certain body positions allow efficient transmission of force and why others lead to slipping feet or tweaked fingers:

- Perpendicular force application to a hold maximises friction and reduces the chance of slipping
- Oblique force on a finger hold increases shear load
- Balanced body positioning distributes load across multiple points of contact, reducing peak force at any single hold
- Placing as much rubber on a sloping volume increases friction and reduces slippage

2.4 Centre of Mass

The **centre of mass** (CoM) is the single point at which the entire mass of a body can be considered to be concentrated for the purposes of analysing motion. It lies roughly at the level of the second sacral vertebra in anatomical position, though it shifts continuously with limb position and can even fall outside the body entirely. What matters for balance is that its vertical projection falls within the **base of support**.

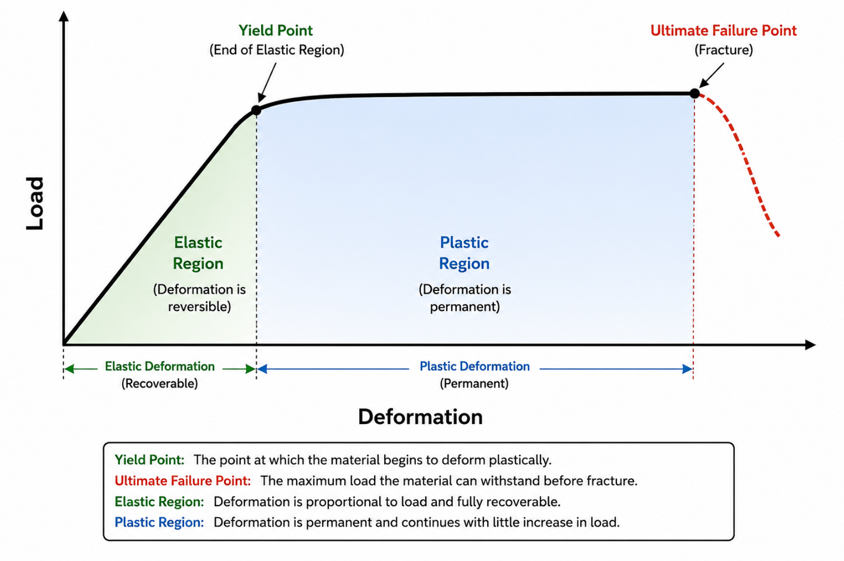
In climbing, a climber who keeps their hips close to the wall and their CoM positioned directly over their feet dramatically reduces the rotational forces pulling them away from the surface, improving both balance and efficiency.



Center of mass positioning close to and far from the climbing wall

2.5 Load Deformation Relationship

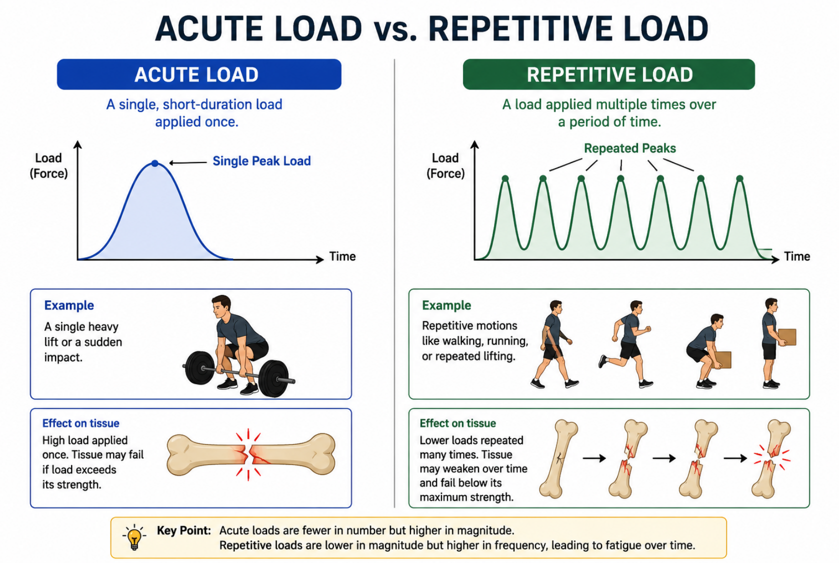
The load vs deformation curve shows how a material responds as force is applied. In the elastic region, the material deforms but returns to its original shape once the load is removed. At the yield point, permanent deformation begins, entering the plastic region. The ultimate failure point is the maximum load the material can withstand before fracturing.



Elastic region, yield point, plastic region, and failure point

Region	What Happens	Clinical Significance
Elastic Region	Tissue returns to original shape when load removed; no permanent damage	Normal training loads should remain in this region
Yield Point	Threshold beyond which permanent deformation begins; microfailure starts	Overuse injuries begin here — cumulative sub-threshold loading
Plastic Region	Permanent structural damage; tissue cannot return to original form	Partial tears; tissue remodelling required
Failure Point	Complete structural failure — fracture or complete rupture	Acute traumatic injury

2.6 Acute vs Repetitive Loading



Single acute load peak versus repeated cyclical loading over time

Load Type	Definition	Example	Clinical Implication
Acute Load	A single force large enough to cause immediate structural damage	Fracture from a fall	Emergency management; structural assessment required
Repetitive Load	Repeated sub-threshold loads; each cycle causes micro-damage that accumulates	Repeated crimping sessions	Stress fractures, pulley overuse, tendinopathies — load management is key

Section 3: Forces, Motion & Muscle Mechanics

Sections 1 and 2 covered planes of motion and lever systems — the geometry of how the body moves. This section adds the physics that explains why those movements behave the way they do.

3.1 Newton's Laws of Motion in Climbing

First Law — Law of Inertia

A body at rest stays at rest, and a body in motion stays in motion, unless acted on by an external force. A climber hanging static on a hold will remain static until a force changes that state — this is why a controlled, deliberate start to a dynamic move is harder than it looks.

Second Law — Force, Mass and Acceleration

Force equals mass multiplied by acceleration (F = m x a). For a given amount of force, a lighter climber will accelerate faster than a heavier one — part of why lighter climbers often find dynamic, momentum-based moves more forgiving.

Third Law — Action and Reaction

Every force a climber applies to a hold is met by an equal and opposite force back into the body. This is why precise, controlled foot placement transmits force efficiently back up through the leg.

COACHING INSIGHT

Newton's laws are rarely taught explicitly to climbers, but coaches use them constantly without naming them. Translating a cue like 'commit to the move' into 'you need to overcome inertia before you can accelerate' can help analytically-minded athletes understand why hesitation costs them on dynamic sequences.

3.2 Describing Movement: Basic Kinematics

Term	Definition	Climbing Application
Distance	Total length of the path travelled, regardless of direction	A climber who readjusts their feet twice covers more distance than displacement — wasted movement costs energy
Displacement	The straight-line distance between start and end points	Route efficiency minimises the gap between distance and displacement — economy of movement
Speed / Velocity	Distance covered per unit of time; velocity includes direction	A static approach has low velocity; a dynamic move has high velocity
Acceleration	The rate of change of velocity over time	Explosive moves require rapid acceleration; lock-offs require smooth deceleration

3.3 Mechanical Advantage: Quantifying 'Stay Close to the Wall'

Mechanical advantage (MA) is the ratio between the muscle's moment arm and the resistance's moment arm:

MA = Moment arm of muscle force / Moment arm of resistance force

An MA greater than 1 means the muscle has a mechanical advantage. An MA less than 1 means the muscle must generate more force than the load itself. Almost every major climbing movement operates at MA below 1, because most human joints are third-class lever systems.

COACHING INSIGHT

This is the mechanism behind 'stay close to the wall.' Moving the hips toward the wall shortens the moment arm of the resistance force (bodyweight), meaning the same muscles need to generate less force to hold the same position. This is precisely why two climbers of equal strength can feel completely different levels of effort on the same hold, based purely on hip position.

3.4 Muscle Mechanics for Coaches

Action	What Happens	Climbing Example
Concentric	The muscle shortens because contraction force exceeds resistance force	Pulling into a lock-off; biceps and lats shorten to draw the body up
Eccentric	The muscle lengthens under load because resistance exceeds contraction force	Lowering with control on a downclimb — higher injury risk as tendons load while stretching
Isometric	Muscle length stays constant; contraction force equals resistance force	Holding a static lock-off or a mid-route rest position

Force-Length Relationship

A muscle produces its greatest force at a mid-range length. This explains why a lock-off becomes noticeably weaker in the last few degrees of full elbow flexion or extension — the muscle is operating outside its strongest length range.

Force-Velocity Relationship

Force and speed of muscle shortening are inversely related: the faster a muscle contracts, the less force it can produce. This is part of why dynamic moves rely on technique and timing rather than pure strength.

COACHING INSIGHT

Together, these two relationships explain why slow strength work (e.g. controlled hangboard repeaters) and fast power work (e.g. campus board, dynos) train genuinely different qualities. Programming for climbing should include both ends of the force-velocity spectrum.

Section 4: Fundamental Movement Patterns

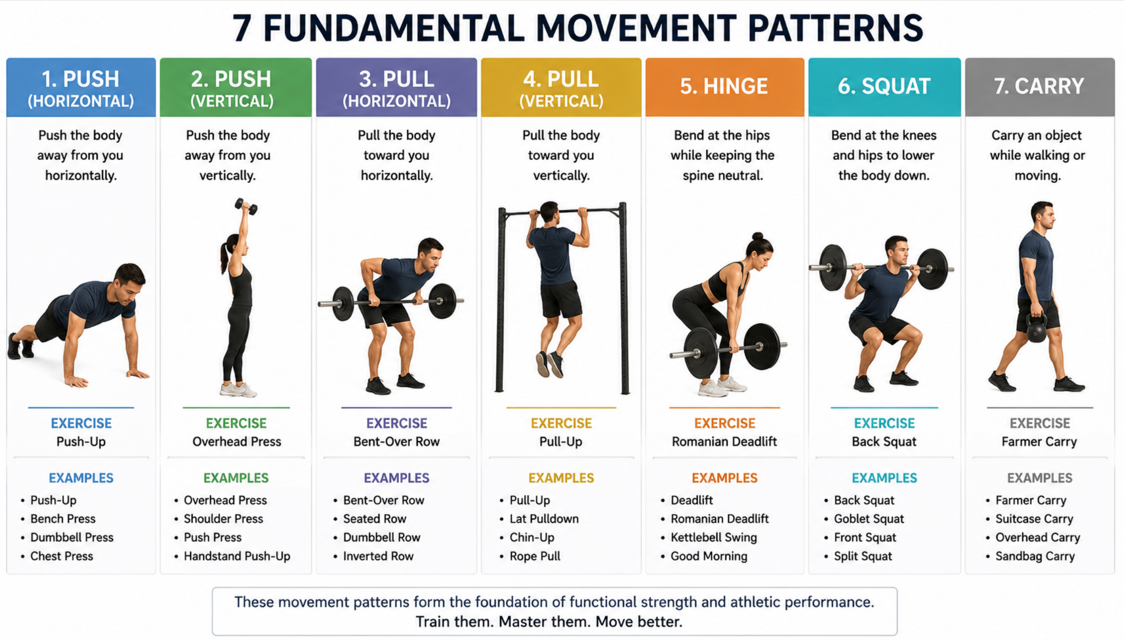
Coaching movement begins with understanding fundamental patterns that form the basis of all human physical activity. In climbing, these patterns manifest on the wall, and their quality directly impacts performance and injury risk.

PROGRAMMING NOTE

A balanced training programme for climbing should include all seven patterns, not just the pulling movements that dominate on the wall. Many common climbing injuries stem directly from an imbalanced training diet that prioritises vertical pulling at the expense of pushing and hip-hinging.

4.1 The Seven Fundamental Patterns

Pattern	Description & Climbing Relevance
Push (Horizontal)	Press-up, dumbbell press. Antagonist to pulling; essential for shoulder health. Mantling is a climbing-specific push.
Push (Vertical)	Overhead pressing — shoulder press. Relevant for shoulder stability during high-reach moves.
Pull (Horizontal)	Bent-over row. Builds scapular retractors and mid-back stability; relevant for pulling into a wall and roof positions.
Pull (Vertical)	Pull-up, lat pull-down. The primary movement pattern in climbing; directly trains lat, biceps, posterior shoulder.
Hinge	Deadlift, Romanian deadlift. Builds posterior chain; trains hip extension for high steps and heel hooks.
Squat	Squat, goblet squat. Strengthens quads and glutes; improves leg drive for standing on high holds and dynos.
Carry	Farmer's carry, suitcase carry. Builds core stability and shoulder girdle resilience.



The seven fundamental movement patterns

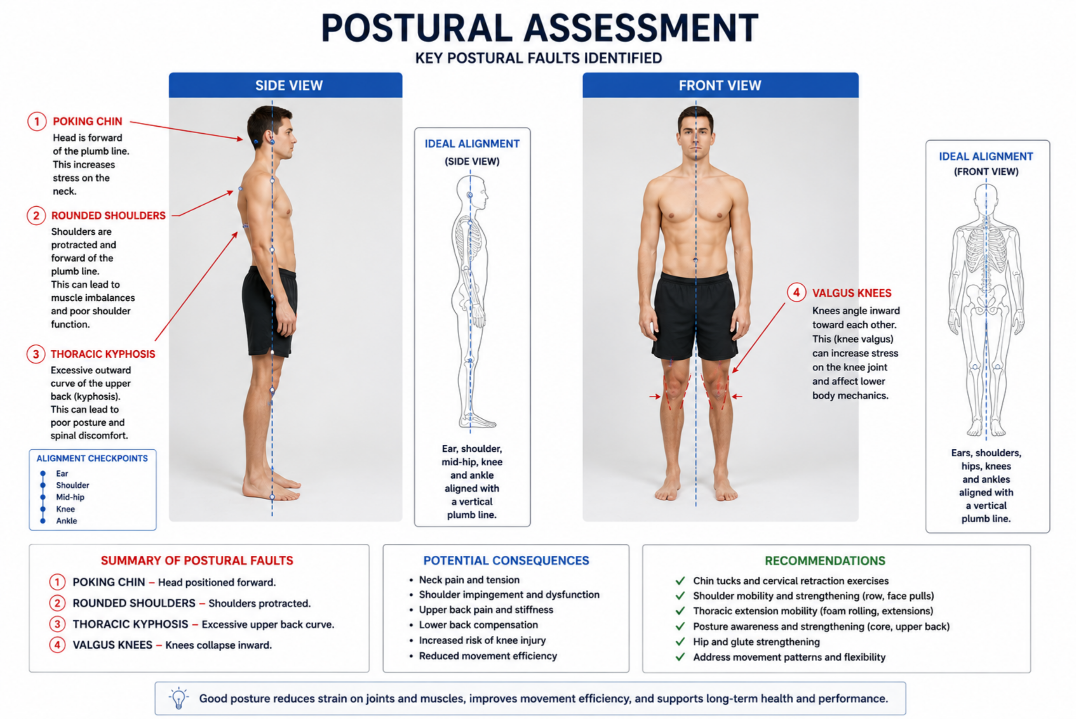
Section 5: Postural Assessment & Movement Dysfunction

Before designing a training programme, coaches should assess their athlete's posture and basic movement quality. This provides a baseline and highlights areas of tightness, weakness, or compensatory movement.

5.1 Static Postural Assessment

A static postural assessment is conducted from the front, side, and rear with the client standing naturally. Common findings in climbers include:

- **Forward head posture** — head protrudes in front of the shoulder line; associated with upper trapezius tightness
- **Rounded shoulders** — associated with tight pectorals and weak rhomboids/lower trapezius
- **Thoracic kyphosis** — excessive rounding of the upper back; increased by repeated overhang climbing
- **Anterior pelvic tilt** — front of the pelvis drops, lower back arches; tight hip flexors, weak glutes/core
- **Knee valgus** — knees cave inward; associated with weak hip abductors and poor proprioception



Common postural faults compared to ideal alignment

IMPORTANT NOTE

Static posture is not diagnostic — it identifies areas to investigate further with dynamic assessment. Many athletes with poor static posture move well under load, and vice versa. Use postural findings as hypotheses, not conclusions.

5.2 Dynamic Movement Assessment — Overhead Squat

The overhead squat is a practical screening tool that simultaneously loads the entire kinetic chain. Observe from the front and side:

Observation	Possible Dysfunction
Arms fall forward	Tight lats or thoracic spine; limited shoulder flexion mobility
Excessive forward lean	Tight hip flexors or ankle dorsiflexion restriction

Observation	Possible Dysfunction
Heels rise	Tight calves / limited ankle mobility
Knees cave inward	Weak hip abductors (glute medius); tight adductors
Low back rounds or arches	Weak core or limited hip mobility

Correct overhead squat bottom position, front and side view

5.3 Mobility vs. Stability — The Joint-by-Joint Approach

Gray Cook's joint-by-joint model proposes that joints alternate in their primary need — some require mobility, others stability. When a mobility-dominant joint is restricted, an adjacent stability-dominant joint compensates.

Joint	Primary Need
Ankle	Mobility (dorsiflexion)
Knee	Stability
Hip	Mobility (multi-directional)
Lumbar spine (low back)	Stability
Thoracic spine (mid back)	Mobility (extension and rotation)
Scapulothoracic	Stability
Glenohumeral (shoulder)	Mobility
Elbow	Stability
Wrist / Hand	Mobility

For climbing coaches: tight hips will cause the lumbar spine to compensate during high steps, placing excessive stress on the lower back. Restricted thoracic mobility causes the shoulder girdle to compensate, increasing impingement risk.

Section 6: Stability, Mobility & Motor Learning

6.1 The Role of Stability in Climbing

- **The core** — transfers and controls forces between upper and lower body, especially on steep terrain
- **The shoulder girdle** — protects the rotator cuff and allows efficient force transmission
- **The lower limb** — maintains precise foot placement and prevents unnecessary body sway

Stability is not about rigidity — it is dynamic. A climber with excellent core stability can move their limbs freely without their torso rotating or sagging — proximal stability enabling distal mobility.

6.2 Mobility Requirements in Climbing

- Hip flexion and external rotation — for high steps, frog positions, and drop-knee technique
- Shoulder flexion and external rotation — for high reaches, underclings, and lock-offs
- Thoracic extension and rotation — for twisting and cross-through moves
- Ankle dorsiflexion — for precise smearing and slab climbing

6.3 Applying Motor Learning in Coaching Practice

Principle	Practical Application
Variability of Practice	Practice similar movements on different holds and angles rather than repeating the same sequence. Builds adaptable motor programmes.
Contextual Interference	Mix different skills within a session rather than blocking by skill type. Initially reduces performance but enhances long-term retention.
Feedback Timing	Delay feedback slightly to encourage self-evaluation. Immediate feedback creates dependency.
Whole vs. Part Practice	Teach the full movement where possible. Only break into parts if the sequence is very long or complex.
Mental Rehearsal	Encourage visualising routes before attempting. Activates similar neural pathways to physical practice.

6.4 Effects of Fatigue

Effect	Mechanism	Coaching Implication
Reduced force output	ATP production rate falls below demand	Reduce load or volume — don't train to failure in technical sessions
Loss of fine motor control	Impaired neuromuscular coordination	Stop technical work when technique breaks down
Increased injury risk	Impaired proprioception and coordination	Monitor technique degradation as a fatigue signal

REHABILITATION IMPLICATION

Training to fatigue must be progressive. Fatigued athletes have impaired neuromuscular control — a primary injury risk factor. Early mobilisation after injury dramatically decreases atrophy and maintains mechanical properties of the musculotendinous unit.

End of Module 02. Return to corelearning.co.za to complete the Module 02 quiz (80% pass mark) and unlock Module 03.

Module Summary

A quick reference before moving on

Key Terms & Definitions

Sagittal Plane	Divides the body into left and right halves; movements include flexion and extension.
Frontal Plane	Divides the body into front and back; movements include abduction, adduction, and lateral flexion.
Transverse Plane	Divides the body into upper and lower halves; movements include rotation (twisting).
Torque	A rotational force; the product of force and the perpendicular distance (moment arm) from the axis of rotation.
Moment Arm	The perpendicular distance between the line of force and the axis of rotation; a longer moment arm increases torque.
Lever System	A rigid structure (bone) rotating around a fulcrum (joint) under effort force (muscle) and resistance (load).
Kinetic Chain	The interconnected system of joints and muscles that work together to produce movement.
Anterior Pelvic Tilt	A postural deviation where the pelvis front drops and the lower back arches; tight hip flexors, weak core.
Thoracic Kyphosis	Excessive rounding of the upper/mid back; commonly seen in climbers due to overhang climbing posture.
Proximal Stability	Stability at the core and shoulder girdle enabling efficient distal (arm/leg) movement.
Joint-by-Joint Model	A framework suggesting joints alternate between needing primarily mobility or stability.
Overhead Squat Assessment	A dynamic movement screen identifying mobility and stability limitations throughout the kinetic chain.
Variability of Practice	A motor learning strategy practicing skill variations rather than exact repetition; improves skill transfer.
Scapular Protraction	Forward movement (rounding) of the shoulder blades; common in climbers due to dominant pulling.

Key Takeaways

- Movement occurs in three planes — sagittal, frontal, and transverse. Climbing is multi-planar and coaches must understand all three.
- Lever mechanics and torque explain why body position dramatically alters the muscular demand of holds and moves. Staying close to the wall reduces moment arms and conserves energy.
- The seven fundamental movement patterns should all feature in a climber's training programme to prevent imbalance.
- Postural assessment identifies common dysfunctions in climbers that can be addressed through targeted mobility and strength work.
- The joint-by-joint model helps coaches understand why restricting mobility at one joint causes compensatory stress elsewhere.
- Motor learning principles — variability, contextual interference, delayed feedback — should inform how coaches structure practice.

LOOKING AHEAD

In Module 3 — Client Assessment & Health Screening — we will apply both anatomical and biomechanical knowledge practically, learning how to assess climbers systematically before designing their programmes.